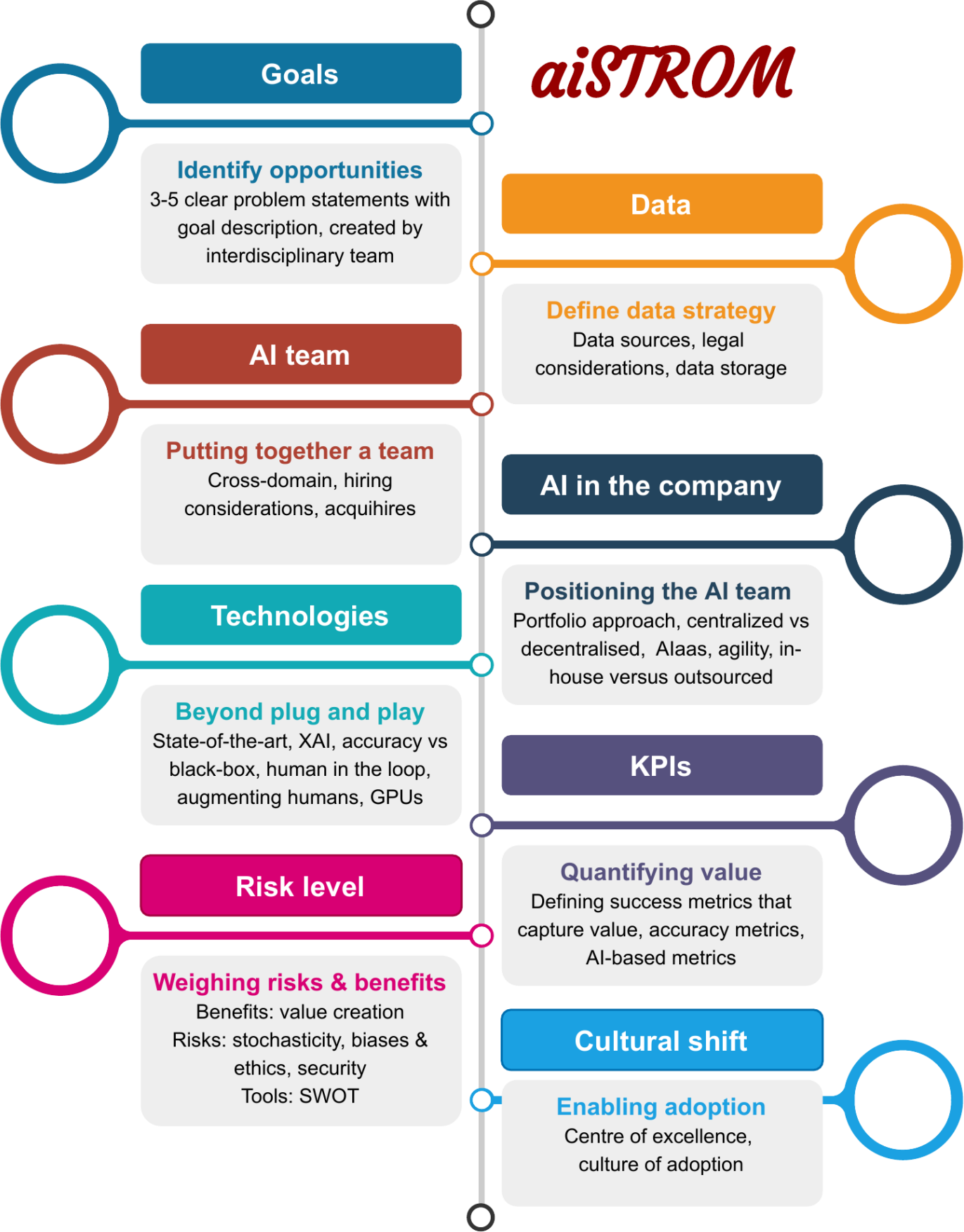


Decide which AI projects to pursue, and what each one requires, before choosing a solution.



How to use these sheets

- 1 Before you start: build an interdisciplinary team**
Combine people with domain and organisation knowledge (e.g. C-level and employees) with people who have AI knowledge (e.g. AI consultants or developers).
- 2 Sheet 1, Goals: identify opportunities**
List possible goals and problem statements that AI could tackle. Select the top n (typically 3-5), maximising potential impact for the lowest effort. Give them letters A, B, C.
- 3 Sheets 2-7: analyse each shortlisted project**
Work through Data, AI team, AI in the company, Technologies, KPIs and Risk level. Each sheet has one column per project, so the differences between projects become visible. End each column with a conclusion.
- 4 Sheet 8, Cultural shift**
Plan the education and adoption the projects need: AI literacy, a centre of excellence, continuous learning.
- 5 Sheet 9, Decide**
Plot the projects by risk and benefit, run a SWOT per project and choose a balanced portfolio that fits the organisation's risk strategy. Uncertain projects usually start with a prototype.

ORGANISATION / STRATEGIC QUESTION

TEAM (DOMAIN AND AI KNOWLEDGE)

STARTING SITUATION

DATE / NEXT REVIEW

Original roadmap, Herremans (2021), Figure 1 (CC BY 4.0)

Based on Herremans, D. (2021). aiSTROM - A roadmap for developing a successful AI strategy. IEEE Access, 9, 155826-155838. Notes marked "2026" are additions.

3-5 clear problem statements with goal description, created by an interdisciplinary team

WHERE TO LOOK FOR IDEAS

Internal efficiency gaps

Where could AI make the organisation perform more smoothly and efficiently?

New AI technologies

Which new technologies exist, and how could they be of use to us?

Competitors

What are competitors doing? What should we emulate, or what are we competing with?

New business models

Are new business models on the rise that could replace our current services?

#	PROBLEM STATEMENT / GOAL DESCRIPTION	HOW AI CREATES VALUE	IMPACT	EFFORT	TOP n
1		-	-	-	-
2		-	-	-	-
3		-	-	-	-
4		-	-	-	-
5		-	-	-	-
6		-	-	-	-
7		-	-	-	-
8		-	-	-	-

AI can create value by 1) automating processing, 2) identifying trends from historic data, or 3) making intelligent predictions that empower people. Choose the top n (typically 3-5) with the highest impact for the lowest effort and give them letters A, B, C; these become the columns of sheets 2-9.

WHY THESE PROJECTS? WHAT HAPPENS TO THE OTHER IDEAS?

Data sources, legal considerations, data storage

	PROJECT A	PROJECT B	PROJECT C
Data sources III-B.1 Does the organisation already have the necessary data? Is big data needed, or will more data not help (curse of dimensionality)? What should we start collecting now?			
Collection vs acquisition III-B.1 Collect it ourselves (time, ethics approval, quality checks), buy it, or annotate it (experts or crowdsourcing)? Garbage in, garbage out.			
Data characteristics III-B Volume, velocity, variety, veracity (bias, noise) and value: what does each mean for this project?			
Legal: privacy & security III-B.2 Which privacy laws apply (e.g. GDPR, PDPA, CCPA; every country we operate in)? Who is the privacy manager? Balance easy internal access against breaches and exfiltration.			
Storing data III-B.3 Internal infrastructure or external data centre? Stored close to its users? Data lake, warehouse or semi-structured? Who administers it?			
Conclusion The data strategy for this project in one or two sentences.			

2026 note: when using hosted generative AI services, prompts and uploaded files are data too. Check where they are stored and whether the provider may train on them.

	PROJECT A	PROJECT B	PROJECT C
Skills needed Mathematics & statistics, data mining, big data engineering, pattern extraction, visualisation & UI, software engineering, communication, continuous learning. Which does this project need?			
Domain knowledge Who brings the domain knowledge into the multidisciplinary team?			
Obtaining talent Up-skill existing employees (technical and vertical AI knowledge), hire (graduates, PhDs, academic collaboration) or acquihire?			
Retaining talent How will we keep AI experts: continuous education, benefits, working environment? Where are we dependent on one person?			
Conclusion The team for this project, and the main gap.			

Portfolio approach, centralised vs decentralised, Alaas, agility, in-house versus outsourced

	PROJECT A	PROJECT B	PROJECT C
Positioning the team Centralised (hub, centre of excellence, chief AI officer), decentralised (spokes) or hub-and-spoke? Does the team have access to data across departments? Short reporting line? Who has decision power?			
Portfolio approach Short term (existing, low-risk technology), medium term (emerging technology) or long term (risky, competitive advantage)? Funded in internal rounds?			
AI as a service (Alaas) Does a service already do this (do not reinvent the wheel)? Is it core technology for us? How exposed are we to price changes?			
In-house, outsource or acquire What is our AI readiness (pioneer, investigator, experimenter, passive)? Build in-house, outsource development, or grow by acquisition?			
Agility Which prototype tests the idea cheaply? Which stakeholders and users to involve early? Will we need MLOps to bring models into production?			
Conclusion Where the project sits and how it will be developed.			

2026 note: Alaas now also means hosted foundation models such as LLMs. The same questions apply: dependence, price changes and what data leaves the organisation.

	PROJECT A	PROJECT B	PROJECT C
State of the art & hybrid Which techniques fit the task? Is it an established task or something nobody has done? Would a hybrid with older AI (rules, fuzzy logic) be more efficient?			
Accuracy vs black box (XAI) Must the outcome be explainable, legally or to the people affected? Would we accept lower accuracy for an interpretable model?			
Human in the loop Where do ground-truth labels come from? Can people add new labels while the system is in production?			
Replacing or augmenting Should the system replace the person doing the task, or augment them?			
Cloud vs in-house hosting What computing (GPU) power is needed? Rent cloud servers or buy equipment (capex and opex)? Small edge devices?			
Conclusion The technical approach for this project.			

2026 note: generative models can produce fluent but wrong output. If a system can act on its own (an 'agent'), decide what it may do without human approval.

Defining success metrics that capture value, accuracy metrics, AI-based metrics

	PROJECT A		PROJECT B		PROJECT C	
Value-based KPIs Which KPIs link the project to strategic goals? Think of customers, costs, processes and investors, not only profit.	III-F					
Hidden value What value is harder to measure: internal efficiency, better understanding of customers, better decisions?	III-F					
Model performance metrics Which accuracy measures (accuracy, confusion matrix, F1) and which level is needed? When would we pivot?	III-F					
AI-based metrics Could AI itself produce better or more insightful KPIs (e.g. sentiment analysis, business intelligence)?	III-F					
Reporting Who reviews the KPIs and how often, to justify continued investment?	III-F					
Conclusion What success looks like, in numbers.						

Benefits: value creation. Risks: stochasticity, biases & ethics, security

	PROJECT A		PROJECT B		PROJECT C	
Stochastic nature III-G.1 There is no guarantee the model will work. How uncertain is feasibility? Do competitors have working models? Is a prototype needed first? What is the pivot?						
Biases & ethics III-G.1 Could historical data teach the model unfair biases? Who could be treated unfairly? Which ethical dilemmas: let the algorithm decide or set rules?						
Security III-G.1 Could deceptive inputs fool the model (adversarial attacks, spoofing)? What else needs protecting?						
Other strategic risks III-G.1 New or hard technology? In-house know-how? Price guarantee for Alaas? Team access to data? A new task or an established one?						
Benefits III-G.2 What value does the project create, for customers or internally?						
Conclusion Overall risk level and benefit level (low / medium / high), for the plot on sheet 9.						

2026 note: for generative AI, add hallucination (stochastic nature) and prompt injection (security).

	PROJECT A	PROJECT B	PROJECT C
Educating managers & team What do the managers and developers directly involved need to know about AI? III-H			
AI literacy for everyone What do users of the system and other employees need to understand? III-H			
From fear to empowerment Do employees fear being replaced? How will we show that AI can empower them, and how they work with the system? III-H; III-E			
Centre of excellence Who creates awareness and knowledge of AI? Can employees join brainstorming workshops about new AI ideas? III-H			
Continuous education How will learning continue as the technology and the system change? III-H			
Conclusion What adoption requires for this project.			

Bring the analysis together and choose a balanced portfolio

RISK-BENEFIT PLOT: write the project letters in the matching quadrant

High value, low risk

e.g. automating complicated decision support

High value, high risk

e.g. highly innovative products and services

Low value, low risk

e.g. automating internal repetitive processes

Low value, high risk

e.g. predicting complex customer behaviour

VALUE (BENEFIT) ->

RISK ->

SWOT PER PROJECT

	A		B		C	
Strengths						
Weaknesses						
Opportunities						
Threats						
Decision	Open		Open		Open	
Next step, owner, date						

ORGANISATION'S RISK STRATEGY

PORTFOLIO DECISION: what we do now, next and later, and how funding is released